SOFTWARE ENGINEER INTERVIEW

Software Engineering & System Simulation

Production Backend Development · GPU LLM Serving Research

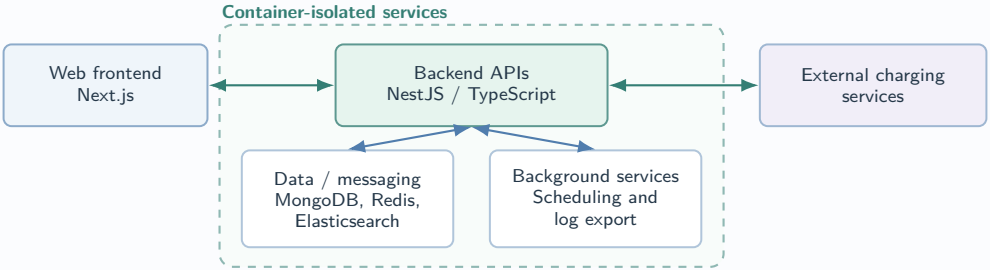
Yu-Jui Liang
M.S. in Data Science
National Taiwan University | Expected Graduation: June 2027
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Backend Integration / System Simulation / Performance Evaluation

NeoPower: Backend System Architecture

01 / PRODUCTION PLATFORM

Backend integration and debugging inside a production EV charging platform.



Backend / API Integration

Developing and integrating services within an existing production system.

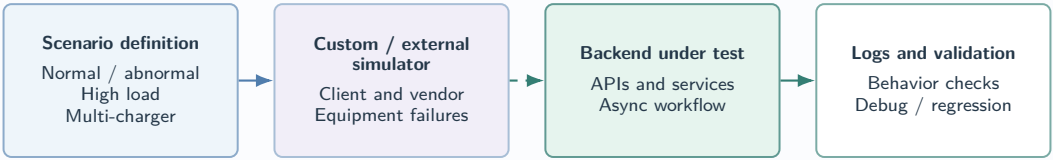
Container-Based Service Isolation

Separating services to support maintenance, deployment, and troubleshooting.

NeoPower: Scenario-Based System Simulation

02 / WHY SIMULATE

Scenarios reproduce charger and vendor behavior when live equipment cannot.



Reproducible workflows

Re-run a defined scenario without the physical charger or live vendor system.

Failure and load

Exercise normal, abnormal, high-load, and multi-charger situations.

Debug and regression

Use logs to investigate bugs and keep past failures from returning.

NeoPower: Testing & Production Reliability

03 / RELIABILITY LOOP

Production reliability needs reproducible tests and feedback from real deployments.

Pre-Deployment Testing

- Scenario-based validation
- Vendor / client simulation
- Regression testing
- High-load and multi-charger failures

Post-Deployment Validation

- Operational log analysis
- Time-based comparison at the same site
- Cross-site comparison, similar conditions
- Software behavior and customer feedback



Field comparisons are observational checks, not randomized experiments. Historical incidents become reusable scenarios for later releases.

From Production Systems to GPU Simulation

04 / SAME ENGINEERING METHOD

Different systems, same loop: model the behavior, then validate it.

NeoPower

Production system simulation for reproducible testing and debugging.

→

GPU LLM Serving Research

System-level simulation for evaluating heterogeneous hardware configurations.

Simulation | **Modeling** | **Validation** | **Performance Analysis**

Shared method across a production backend and current graduate research.

LLM Serving: Heterogeneous GPU & Memory

05 / GRADUATE RESEARCH QUESTION

How should LLM requests be scheduled across GPUs with different compute and memory characteristics?

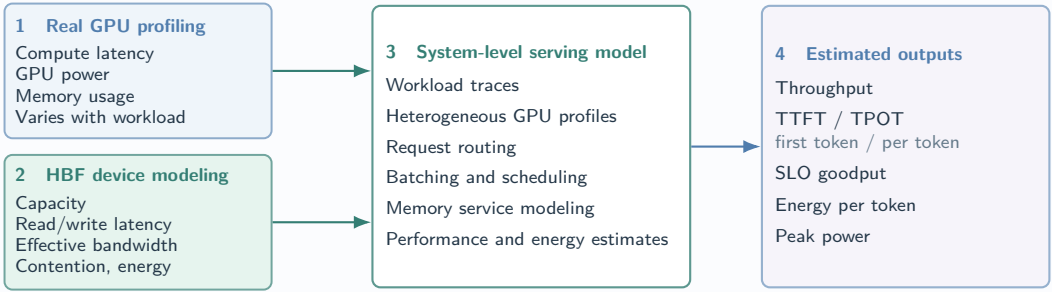
Prefill compute-heavy	**Decode** memory-sensitive	**Model weights** large, read-mostly	**KV cache** latency-sensitive
	HBM-only fast memory	**HBF-dominant** high capacity	**Hybrid**
High compute	Latency-first memory	Capacity-first may need buffering	HBM + HBF split by role
Low compute	Latency-first memory	Capacity-first may need buffering	HBM + HBF split by role

HBF in this study. Higher capacity, with different latency, bandwidth, power, and endurance. The six cells are conceptual simulation profiles, not commercial GPU products.

Research Methodology: Simulation Framework

06 / PROPOSED METHOD

Evaluate routing and energy with a system-level serving model.



Candidates, not yet integrated: LLMServingSim 2.0 and FlashLab HBFSim. Scope is system-level behavior, including memory-service contention, not GPU cycle simulation.

Current Progress & Research Roadmap

07 / ONGOING RESEARCH

The question and method are in place. Integration and results are still ahead.

Current Progress

Completed

✓ Literature survey on heterogeneous GPU serving and HBF memory systems.

✓ Initial formulation of the heterogeneous compute–memory design space.

✓ Evaluation of candidate simulation frameworks and the research method.

Next Steps

Planned

→ Reproduce LLM serving simulator baselines.

→ Profile real GPU computation and energy.

→ Integrate HBM/HBF memory models.

→ Explore HBF-aware request routing.

→ Evaluate performance–energy trade-offs under latency SLOs.

Thank You

Questions & Discussion